Melih Barsbey

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SUMMARY

AI researcher working on efficient and trustworthy machine learning. Current research focuses on the safety and reliability of LLMs and VLMs under resource constraints, and how representation geometry mediates the effects of these constraints. Research published in *NeurIPS*, *ICML*, *ICLR*, *ICCV*, and *Nature Medicine* among others; Area Chair for *ICLR*. Further research topics include mechanistic interpretability, learning theory, and medical decision-making with human-AI complementarity and noisy supervision. Industry experience in applied ML research, deployment, and team leadership, including daily demand forecasting across 8,000+ ATMs.

PROFESSIONAL EXPERIENCE

Imperial College London, London, United Kingdom

- Postdoctoral Research Associate in the Department of Computing — Jul 2024 – Present
  - Safety and alignment of LLMs and VLMs under quantization, pruning, and low-rank compression; jailbreak and prompt-injection evaluation.
  - Inference reliability and post-training stability under weight, activation, and KV cache quantization; evaluation and mitigation strategies.
  - Reliability of transformer models under resource-efficient design: sparse attention, mixture-of-experts, multi-head latent attention.
  - Additional published research on mechanistic interpretability, topological deep learning for drug discovery, robustness to spurious correlations, grokking and learning theory.

Google DeepMind, London, United Kingdom

- Research Scientist Intern, Robust and Verified AI team — Jun 2022 – Oct 2022
  - Development of a confidence-based deferral system for robust human-AI decision-making with sparse medical data.
  - Evaluation and training of models under noisy, conflicting annotations: ranking models over partially ordered expert judgments; the k-wise generalization of the Bradley-Terry preference model.

Freelance Machine Learning Consultant

- Freelance consultancy for multiple software companies on the research, development, and deployment of machine learning systems. — Jul 2019 – Jun 2024
  - Development of AutoML pipelines for demand forecasting.
  - System design for ML-based segmentation of e-commerce customers.

Arute Solutions, Istanbul, Turkey

- Senior ML Scientist, Team Lead — Sep 2017 – Jun 2019
  - Leadership of a team of five ML scientists and engineers developing the flagship AI product: cash-demand forecasting deployed in production across 8,000+ ATMs.

EDUCATION

Boğaziçi University, Istanbul, Turkey

- PhD in Computer Engineering — Sep 2016 – Sep 2023
  - Focus: Latent variable models, compressibility and generalization
  - Advisor: Prof. A. Taylan Cemgil
  - Cumulative GPA: 4.00 / 4.00
- B.A., M.A. in Psychology — Sep 2007 – Aug 2014
  - Focus: Embodied language processing, causal explanations
  - Thesis: The effect of narrative shifts on perceptual simulations

SELECTED PUBLICATIONS AND MANUSCRIPTS

[14] **Barsbey, M.**, Prieto, L., Stevinson, E., & Birdal, T. Reliability under Pressure: How Quantization Affects Safety, Inference Robustness, and Post-Training Stability Through Layerwise Error Propagation. *In preparation*, 2026.

[13] **Barsbey, M.**, Ribeiro, A. H., Şimşekli, U., & Birdal, T. On the Interaction of Compressibility and Adversarial Robustness. *International Conference on Learning Representations (ICLR)*, 2026.

[12] **Barsbey, M.**, Stevinson, E., Prieto, L., & Birdal, T. Vulnerability to Multimodal Jailbreak and Prompt Injection Attacks under Activation Quantization. *ECCV Workshop on Universal Representations for Perception, Reasoning, and World Modeling*, 2026.

[11] Stevinson, E., Prieto, L., **Barsbey, M.**, & Birdal, T. Adversarial Vulnerability from Interference Between Features in Superposition. *International Conference on Machine Learning (ICML)*, 2026.

[10] Prieto, L., Stevinson, E., **Barsbey, M.**, Birdal, T., & Mediano, P. A. M. Correlations in the Data Lead to Semantically Rich Feature Geometry Under Superposition. *International Conference on Learning Representations (ICLR)*, 2026.

[9] **Barsbey, M.**, Prieto, L., Zafeiriou, S., & Birdal, T. Large Learning Rates Simultaneously Achieve Robustness to Spurious Correlations and Compressibility. *International Conference on Computer Vision (ICCV)*, 2025.

[8] Prieto, L., **Barsbey, M.**, Mediano, P. A. M., & Birdal, T. Grokking at the Edge of Numerical Stability. *International Conference on Learning Representations (ICLR)*, 2025.

[7] Stutz, D.[*], Cemgil, A. T.[*], Roy, A. G.[*], Matejovicova, T.[*], **Barsbey, M.**[*], et al. Evaluating AI Systems under Uncertain Ground Truth: A Case Study in Dermatology. *Medical Image Analysis, 103*, p.103556, 2025. [*]Equal contribution.

[6] Wan, Y.[*], **Barsbey, M.**[*], Zaidi, A., & Şimşekli, U. Implicit Compressibility of Overparametrized Neural Networks Trained with Heavy-Tailed SGD. *International Conference on Machine Learning (ICML)*, 2024. [*]Equal contribution.

[5] Dvijotham, K.[*], Winkens, J.[*], **Barsbey, M.**[*], Ghaisas, S.[*], Pawlowski, N. et al. Enhancing the reliability and accuracy of AI-enabled diagnosis via complementarity-driven deferral to clinicians (CoDoC). *Nature Medicine*, 29 (7), 1814-1820, 2023. [*]Equal contribution.

[4] **Barsbey, M.**, & Cemgil, A. T. Modeling Hierarchical Seasonality through Low-Rank Tensor Decompositions in Time Series Analysis. *IEEE Access*, 11, 85770-85784, 2023.

[3] **Barsbey, M.**[*], Özçelik, R.[*], Bağ, A. et al. pydebiaseddta: A Computational Software for Training Robust Drug-Target Affinity Prediction Models. *Journal of Computational Biology*, 30 (11), 1240-1245, 2023. [*]Equal contribution.

[2] Raj, A., **Barsbey, M.**, Gürbüzbalaban, M., et al. Algorithmic Stability of Heavy-Tailed Stochastic Gradient Descent on Least Squares. *Conference on Algorithmic Learning Theory (ALT)*, 2023.

[1] **Barsbey, M.**[*], Sefidgaran, M.[*], Erdogdu, M. A., Richard, G., & Şimşekli, U. Heavy Tails in SGD and Compressibility of Overparametrized Neural Networks. *Advances in Neural Information Processing Systems Conference (NeurIPS)*, 2021. [*]Equal contribution.

TECHNICAL SKILLS

Languages & frameworks	Python, Julia; PyTorch, JAX
Training & inference	PyTorch Lightning, HF Transformers, TRL, PEFT, vLLM
Engineering	Distributed training, SLURM, Linux, Docker, Git, pytest
Safety evaluation	HarmBench, JailbreakBench, lm-eval, LLM-as-a-judge
Experiment tooling	Hydra, Weights & Biases, MLflow
Compression	bitsandbytes, torchao, torch-pruning; low-precision formats (e.g. INT8, FP8, NVFP4)

SCIENTIFIC SERVICE

Area Chair

- International Conference on Learning Representations (ICLR) — 2026 – Present

Reviewer

- Neural Information Processing Systems (NeurIPS) — 2019 – Present
- International Conference on Machine Learning (ICML) — 2021 – Present
- International Conference on Learning Representations (ICLR) — 2021 – Present
- Conference on Computer Vision and Pattern Recognition (CVPR) — 2024 – Present
- AAAI Conference on Artificial Intelligence (AAAI) — 2026 – Present

RECOGNITION & SCHOLARSHIPS

- UK Global Talent Visa, endorsed by UKRI — 2024 – 2029
- PhD Scholarship from The Scientific and Technological Research Council of Turkey (TÜBİTAK) — 2016 – 2021

SUPERVISION & MENTORING

- Co-supervision of two PhD students in mechanistic interpretability and safety, Imperial College London, with Prof. Tolga Birdal — 2024 – Present
- Team lead for five ML scientists and engineers at Arute Solutions: hiring, onboarding, technical mentoring, and project direction — 2017 – 2019